適用於支援功能區的災難恢復FlexGroup

ONTAP 9

NetApp
May 11, 2022

This PDF was generated from https://docs.netapp.com/zh-tw/ontap/flexgroup/disaster-recovery-workflow-task.html on May 11, 2022. Always check docs.netapp.com for the latest.

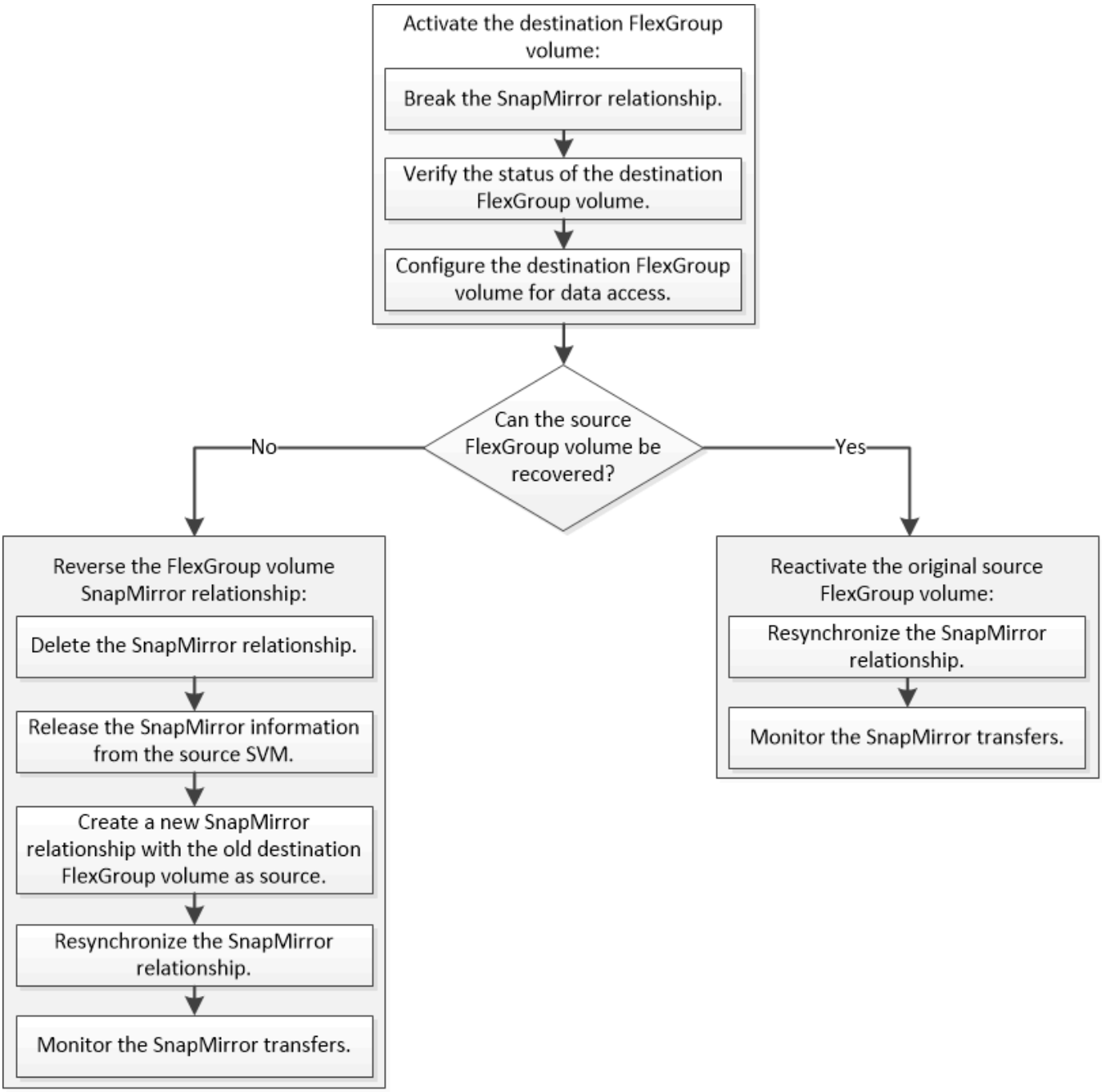
目錄

適用於支援功能區的災難恢復FlexGroup . . . 1
適用於支援功能區的災難恢復工作流程FlexGroup . . . 1
啟動目的地FlexGroup 的目的地功能 . . . 2
災難發生後重新啟動原始來源FlexGroup 的s功能 . . . 4
在FlexGroup 災難恢復期間、反轉SnapMirror在各個方面的關係 . . . 5

適用於支援功能區的災難恢復FlexGroup

適用於支援功能區的災難恢復工作流程FlexGroup

當災難發生在來源FlexGroup 的供應區時、您應該啟動目的地FlexGroup 的供應區、並重新導向用戶端存取。視來源FlexGroup 的來源供應區是否可以恢復而定、您應該重新啟動來源FlexGroup 供應區、或是反轉SnapMirror關係。



當FlexGroup 某些SnapMirror作業（例如SnapMirror中斷和重新同步）正在執行時、用戶端對目的地的存取功能會在短時間內遭到封鎖。如果SnapMirror作業失敗、部分成分可能仍處於這種狀態、FlexGroup 無法存取到此磁碟區。在此情況下、您必須重試SnapMirror作業。

啟動目的地FlexGroup 的目的地功能

當來源FlexGroup 供應資料時、由於資料毀損、意外刪除或離線狀態等事件而無法提供資料時、您必須啟動目的地FlexGroup 的穩定區才能存取資料、直到您在來源FlexGroup 的穩定區上恢復資料為止。啟動作業包括停止未來的SnapMirror資料傳輸、以及打破SnapMirror關係。

您必須從目的地叢集執行此工作。

步驟

1. 停用FlexGroup 針對Sesfvolume SnapMirror關係的未來傳輸：「napmirror quiesce dest_Svm:dest_flexgroup」（napmirror靜止dest_Svm:dest_flexgroup）

```
cluster2::> snapmirror quiesce -destination-path vsd:dst
```

2. 打破FlexGroup 「SnapMirror磁碟區」關係：「napmirror Break dest_svm:dest_flexgroup」

```
cluster2::> snapmirror break -destination-path vsd:dst
```

3. 檢視SnapMirror關係的狀態：「napmirror show -expand」

```
cluster2::> snapmirror show -expand

Progress
Source            Destination Mirror  Relationship   Total
Last
Path        Type  Path        State   Status         Progress  Healthy
Updated
----------- ---- ------------ ------- -------------- --------- -------
--------
vss:s       XDP  vsd:dst        Broken-off
                                    Idle           -         true    -
vss:s__0001 XDP  vsd:dst__0001  Broken-off
                                    Idle           -         true    -
vss:s__0002 XDP  vsd:dst__0002  Broken-off
                                    Idle           -         true    -
vss:s__0003 XDP  vsd:dst__0003  Broken-off
                                    Idle           -         true    -
vss:s__0004 XDP  vsd:dst__0004  Broken-off
                                    Idle           -         true    -
vss:s__0005 XDP  vsd:dst__0005  Broken-off
                                    Idle           -         true    -
vss:s__0006 XDP  vsd:dst__0006  Broken-off
                                    Idle           -         true    -
vss:s__0007 XDP  vsd:dst__0007  Broken-off
                                    Idle           -         true    -
vss:s__0008 XDP  vsd:dst__0008  Broken-off
                                    Idle           -         true    -
...
```

每個組織的SnapMirror關係狀態都是「中斷」。

4. 確認目的地FlexGroup 的Svm_name為讀取/寫入：「volume show -vserver Svm_name」

```
cluster2::> volume show -vserver vsd
Vserver   Volume       Aggregate    State      Type       Size
Available Used%
--------- ------------ ------------ ---------- ---- ----------
---------- -----
vsd       dst          -            online     **RW**          2GB
1.54GB   22%
vsd       d2           -            online     DP          2GB
1.55GB   22%
vsd       root_vs0     aggr1        online     RW        100MB
94.02MB    5%
3 entries were displayed.
```

5. 將用戶端重新導向至目的地FlexGroup 的不確定磁碟區。

災難發生後重新啟動原始來源FlexGroup 的s功能

當來源FlexGroup 供應時、您可以重新同步原始來源FlexGroup 和原始目的地的資料。目的地FlexGroup 的任何新資料都會遺失。

目的地磁碟區上的任何作用中配額規則都會停用、而且配額規則會在執行重新同步之前刪除。

您可以使用「Volume配額原則規則create」和「volume quota modify」命令、在重新同步作業完成後建立和重新啟動配額規則。

步驟

1. 從目的地叢集、重新同步FlexGroup 處理Sfestvolume SnapMirror關係：「napmirror resSync -dest-path dst_svm:dest_flexgroup」
2. 檢視SnapMirror關係的狀態：「napmirror show -expand」

```
cluster2::> snapmirror show -expand

Progress
Source            Destination Mirror  Relationship   Total
Last
Path        Type  Path        State   Status         Progress  Healthy
Updated
----------- ---- ------------ ------- -------------- --------- -------
--------
vss:s       XDP  vsd:dst        Snapmirrored
                                      Idle           -         true    -
vss:s__0001 XDP  vsd:dst__0001  Snapmirrored
                                      Idle           -         true    -
vss:s__0002 XDP  vsd:dst__0002  Snapmirrored
                                      Idle           -         true    -
vss:s__0003 XDP  vsd:dst__0003  Snapmirrored
                                      Idle           -         true    -
vss:s__0004 XDP  vsd:dst__0004  Snapmirrored
                                      Idle           -         true    -
vss:s__0005 XDP  vsd:dst__0005  Snapmirrored
                                      Idle           -         true    -
vss:s__0006 XDP  vsd:dst__0006  Snapmirrored
                                      Idle           -         true    -
vss:s__0007 XDP  vsd:dst__0007  Snapmirrored
                                      Idle           -         true    -
vss:s__0008 XDP  vsd:dst__0008  Snapmirrored
                                      Idle           -         true    -
...
```

每個組織的SnapMirror關係狀態都是「napMirror鏡 射」。

在FlexGroup 災難恢復期間、反轉SnapMirror在各個方面的關係

當災難停用FlexGroup SnapMirror關係的來源等量資料時、您可以在FlexGroup 修復或更換來源FlexGroup 的等量資料時、使用目的地的等量資料。在來源FlexGroup 供應資料的來源供應上線後、您可以將原始來源FlexGroup 供應資料的來源供應區變成唯讀目的地、並反轉SnapMirror關係。

目的地磁碟區上的任何作用中配額規則都會停用、而且配額規則會在執行重新同步之前刪除。

您可以使用「Volume配額原則規則create」和「volume quota modify」命令、在重新同步作業完成後建立和重新啟動配額規則。

步驟

1. 在原始目的地FlexGroup 的Sfetsvolume上、移除來源FlexGroup 供應區與目的地FlexGroup 供應區之間的資料保護鏡像關係：「napmirror DELETE -destination-path Svm_name：volume名稱」

```
cluster2::> snapmirror delete -destination-path vsd:dst
```

2. 在原始來源FlexGroup 的Sfidvolume上、從來源FlexGroup 的Sfidvolume移除關係資訊：「napmirror release -destination-path Svm_name：volume名稱-bidation-info-only」

   刪除SnapMirror關係之後、您必須先從來源FlexGroup 的SnapMirror Volume移除關係資訊、然後再嘗試重新同步作業。

```
cluster1::> snapmirror release -destination-path vsd:dst -relationship
-info-only true
```

3. 在新的目的地FlexGroup Sfidvolume上、建立鏡射關係：「napmirror create -source-path sr_svm_name：volume名稱-dite-path dst_svm_name：volume名稱-type XDP -policy MirrorAllSnapshots」

```
cluster1::> snapmirror create -source-path vsd:dst -destination-path
vss:src -type XDP -policy MirrorAllSnapshots
```

4. 在新的目的地FlexGroup Sesfvolume上、重新同步來源FlexGroup Sesf:「napmirror resSync -source-path Svm_name：volume名稱」

```
cluster1::> snapmirror resync -source-path vsd:dst
```

5. 監控SnapMirror傳輸：「napmirror show -expand」

```
cluster2::> snapmirror show -expand

Progress
Source            Destination Mirror  Relationship   Total
Last
Path        Type  Path        State   Status         Progress  Healthy
Updated
----------- ---- ------------ ------- -------------- --------- -------
--------
vsd:dst       XDP  vss:src        Snapmirrored
                                       Idle           -         true    -
vss:dst__0001 XDP  vss:src__0001  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0002 XDP  vss:src__0002  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0003 XDP  vss:src__0003  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0004 XDP  vss:src__0004  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0005 XDP  vss:src__0005  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0006 XDP  vss:src__0006  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0007 XDP  vss:src__0007  Snapmirrored
                                       Idle           -         true    -
vsd:dst__0008 XDP  vss:src__0008  Snapmirrored
                                       Idle           -         true    -
...
```

每個組織的SnapMirror關係狀態顯示為「napMirroring」、表示重新同步已成功。

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.